Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/03/2019
PI Name:	Cheng, Zhengdong Ph.D.		
Project Title:	Liquid Crystals of Nanoplates		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Liquid Crystal Facility NOTE: Extended to 8/31/2020 per NSSC information (Ed., 2/14/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AQ60G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The central objectives of the proposed research are to investigate the dynamics of phase transitions, including the formation of liquid crystals as well as the equilibrium gel or empty liquid state, of highly anisotropic colloidal disks. For the FY2019, ground based research was focused on the phase transition of discotic liquid crystal suspensions in the presence of an electric field, as well as the effect of temperature gradient on the nucleation, growth, and dispersion of crystals within a channel.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The understanding of the structure and properties of colloidal disks are important to soil science and space exploration (e.g., Mars), to the fundamental investigation of discotic liquid crystals, as well as various industries, including oil recovery, the delivery of pharmaceuticals, nutrition, and cosmetics. The investigated liquid crystals of colloidal platelets are very intriguing from the fundamental science perspective. Besides, external field induced alignment of liquid crystal nanoparticles may find applications in optical devices.
Task Progress:	The phase behavior of nanoparticles is largely dependent on the composition and concentration of particles in the system itself. Furthermore, phase behavior is largely affected by external field such as temperature, magnetic fields, electric fields, and flow fields. External fields have yielded unique mesophase behaviors, as well as better means of control on the dynamics of phase transition and nanoparticle alignment, otherwise conventionally elusive. Herein, we focus on liquid crystal phase transition kinetics and the effects of external electric field and temperature gradient on nanoplate liquid crystals. Previous works investigated nucleation and defect splitting in liquid crystals under the application of electric field as well as the relationship between particle geometry and the Kerr coefficient. We found that the Kerr coefficient and the anisotropy of polarizability varies directly proportionally with the aspect ratio of the particles. The I-S transition in discotics occurred via nucleation and growth mechanism. While polymer grafted nanoplates, above a certain concentration, showed iridescence due to lamellar order of nanoplates. We also demonstrated that application of the electric field yielded a biaxial phase by rotation and splitting of defect brushes around the singularity point. Lastly, thermos-diffusion experiment on nanoplate suspensions showed that nanoplates diffuse towards the higher temperature zone. Current work performed focused on the phase transitions occurring within discotic liquid crystals in the presence of an external electric field. Zirconium Phosphase nanoplatelets of negative anisotropic susceptibility were produced by hydrothermal synthesis in volume fractions ranging from 0.006 to 0.015. We experimentally demonstrated the formation of the biaxial nematic phase in the presence of an electric field in negatively anisotropic discotic crystal systems. The underlying reasoning for the transition to nematic phase was observed to be the splitting of ±1 defects and the re-alignment of domains into higher order and smaller sizes. The phase diagram in the electric field strength - concentration plane reveals the existence of a tricritical point. This tricritical point occurred at the point of convergence between a second order transition and a first order transition. Above this tri-critical point, the biaxial transition curve was observed to be of higher order than 1 for this system, and a threshold volume fraction of about $\varphi = 0.0007 \pm 0.0001$ was observed below which no phase transition occurred. We also demonstrate that external temperature gradients can be effective in controlling nematic LC order in nanoplate colloids. Temperature gradient can be defined as the rate of change of temperature under a given arrangement direction. Two-dimensional zirconium phosphate (ZrP) nanoparticles were used as the template for investigation of liquid crystal phase behavior. ZrP nanoplatelets were synthesized by hydrothermal and reflux methods and its size and height were tuned to form liquid crystals as needed for the study. The isotropic and isotropic-nematic phases of ZrP was preliminarily studied in the sealed capillary. The initial isotropic phase was aligned and transformed to a nematic phase with temperature increment and the motion of ZrP nanoplatelets liquid crystals was clearly observed under a polarizer. This was due to a net density gradient established in the isotropic suspension of nanoplates by a temperature gradient, thus entropically favoring the formation of nematic LC phase in hot regions as a result of nanoplate crowding. Temperature gradient driven growth of nematic phase featured large nematic monodomains. Absence of hedgehog defects indicated more stable structures owing mostly to slow growth of LC phase controlled by external temperature gradient than that formed spontaneously via nucleation and growth of a sufficiently high concentration suspension of nanoplates. Due to their high surface area-to-volume ratio and excellent thermophoretic properties, nanoplates can prove to be ideal candidates for transport of biomolecules through temperature varying environments. Controlling colloidal self-assemblies using external forces is essential to engineer modern electro-optical and biomedical devices. In particular, liquid crystal (LC) assembly of shape anisotropic colloids such as nanoplates can provide control on unique optical properties such as birefringence.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	Huang D, Zeng M, Wang L, Zhang L, Cheng Z. "Biomimetic colloidal photonic crystals by coassembly of polystyrene nanoparticles and graphene quantum dots." RSC Advances. 2018 Oct;8(61):34839-47. https://doi.org/10.1039/C8RA07158F , Oct-2018
Articles in Peer-reviewed Journals	Zeng M, Huang D, Wang P, King D, Peng B, Luo J, Lei Q, Zhang L, Wang L, Shinde A, Shuai M, Clark N, Cheng Z. "Autonomous catalytic nanomotors based on 2D magnetic nanoplates." ACS Applied Nanomaterials. 2019 Jan;2(3):1267-73. https://doi.org/10.1021/acsanm.8b02153 , Jan-2019